



DEPARTMENT OF COMPUTER SCIENCE - GRADUATION PROJECT

Voice Fraud Detection System Using *Artificial Intelligence*

An AI-Powered Multimodal Framework for Real-Time Arabic Telecommunication Security



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1. INTRODUCTION

Voice-based fraud is rapidly evolving as fraudsters exploit AI-generated voices, social engineering scripts, and operator-hopping tactics to deceive victims and cause financial losses. Current detection methods often struggle under real telecommunication conditions, including short audio clips, background noise, and Arabic dialect variations. This project presents SOUN, a multimodal AI-based voice fraud detection system for Arabic telecommunication environments. As one of the first systems of its kind in the Arab region, SOUN supports Arabic with high accuracy and analyzes calls in real time through three dimensions: what is being said, who is saying it, and whether the voice is authentic, classifying each call into four risk levels: No Fraud, Low, Medium, and High

2. Methodology & System Architecture

Incoming audio passes through an Audio Ingestion & Preprocessing stage that segments it into duration-specific chunks. The chunks are dispatched in parallel to three independent detection modules, each producing a binary output. The Risk Aggregation & Classification component fuses these signals and assigns one of four risk levels (No fraud · Low · Medium · High) returned through a FastAPI backend.

01 Speech Pattern Analysis ALLAM-7B · QLoRA
Fine-tuned Arabic LLM classifies the transcribed call content as fraudulent or normal, capturing wording, phrasing, and conversational cues.

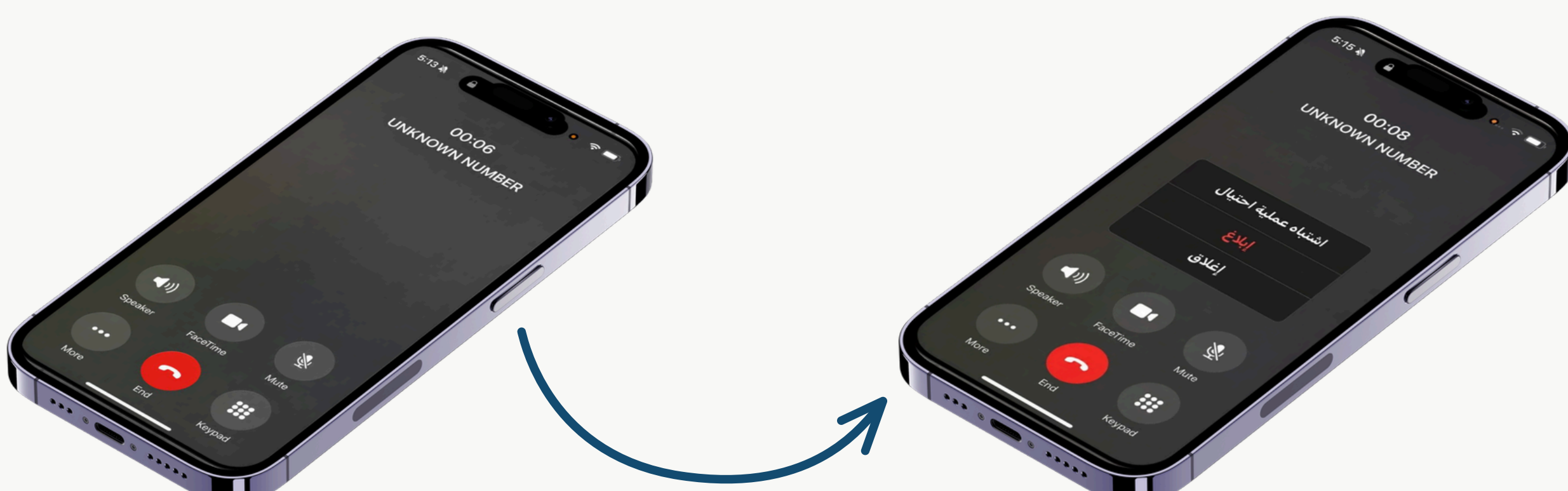
02 Voiceprint Recognition & Matching ECAPA-TDNN · FT
Speaker-embedding model compares the caller's voiceprint against a database of known fraudsters via cosine similarity, even on 2-second utterances.

03 Deepfake Audio Detection Catch-the-AI · 2-stage
Two-stage fine-tuned acoustic model identifies AI-generated or manipulated voices, trained against ChatGPT, ElevenLabs, Sora, Lahjati, and HuggingFace TTS.

SYSTEM FRAMEWORK - REAL-TIME PIPELINE



Fraud Call Detection Notification



3. Experiments & Results

Speech Pattern Analysis

DATASET
100k+

99.67 % Accuracy

Baseline ~50 %

Fine-tuned ~99 %

Voiceprint Recognition & Matching

DATASET
20K+

~98 % Accuracy

Baseline ~90 %

Fine-tuned ~98 %

Deepfake Audio Detection

DATASET
25K+

97.57 % Accuracy

Baseline ~22 %

Fine-tuned ~97 %

4. Achivements



JADATHON · 1ST PLACE WINNER



Awarded first place under the sponsorship of JADA30 and the Social Development Bank, recognizing innovation and societal impact in fraud prevention



HUAWEI ICT COMPETITION 2025-2026

Selected to represent Qassim University in the Middle East & Central Asia Innovation Competition for the global competition.



DEPARTMENTAL NOMINATION

Nominated by the Department of Computer Science as one of the top graduation projects to be officially showcased.

5. Conclusion

SOUN integrates linguistic, biometric, and synthetic-voice analysis into one real-time AI pipeline. By fine-tuning on Arabic telecommunication data, it reliably detects voice fraud in short, noisy, and dialect-heavy calls, making it suitable for real-world telecom deployment.